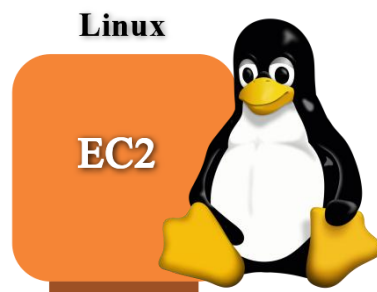




AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

**Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator**

Create an EC2 Linux Instance in AWS



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Perundurai.



Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- **Web and Application Hosting:** Run Apache web servers or other Linux-based web applications in the cloud.
- **Database Servers:** Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- **Remote Desktop Applications:** Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- **Development and Testing:** Create development environments with linux OS to test php, MySql or other Linux-based applications.

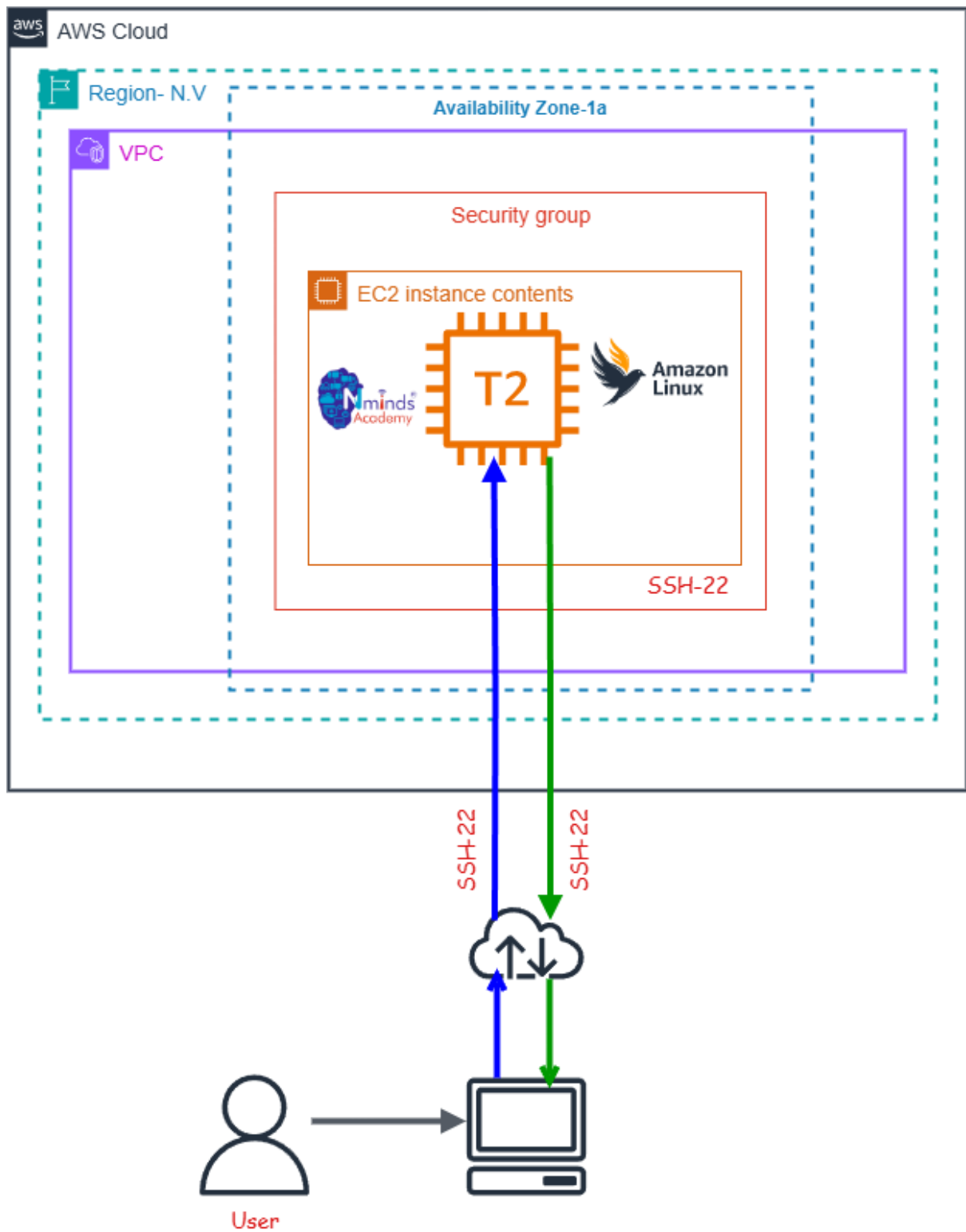
Used Services in AWS



Amazon EC2



Topology



Execution Tasks:

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The page is divided into several sections:

- Name and tags:** A text box with 'linux' and a link to 'Add additional tags'.
- Application and OS Images (Amazon Machine Image):** A search bar and a grid of AMIs. The 'Amazon Linux' AMI is selected.
- Summary:** A sidebar on the right showing the configuration: 1 instance, Amazon Linux 2023 AMI, t3.micro instance type, New security group, and 1 volume (8 GiB).

The screenshot shows the 'Next Steps' page in the AWS Management Console. It features a green success banner at the top and several recommended actions:

- Create billing usage alerts:** To manage costs and avoid surprise bills.
- Connect to your instance:** Once the instance is running, log into it from your local computer.
- Connect an RDS database:** Configure the connection between an EC2 instance and a database.
- Create EBS snapshot policy:** Create a policy that automates the creation, retention, and deletion of EBS snapshots.
- Manage detailed monitoring:** Enable or disable detailed monitoring for the instance.
- Create Load Balancer:** Create an application, network gateway or classic Elastic Load Balancer.
- Create AWS budget:** AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
- Manage CloudWatch alarms:** Create or update Amazon CloudWatch alarms for the instance.

The screenshot shows the 'Instances' page in the AWS Management Console. It displays a table of instances and a detailed view of a specific instance.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
linux	i-071cf392ee5414cd4	Running	t3.micro	Initializing	View alarms	us-east-1d	ec2-54-209-104-139.co...	54.209

i-071cf392ee5414cd4 (linux)

- Instance summary:** Instance ID, IPv6 address, Hostname type, Public IPv4 address, Private IPv4 addresses, Public DNS, Private IP DNS name.





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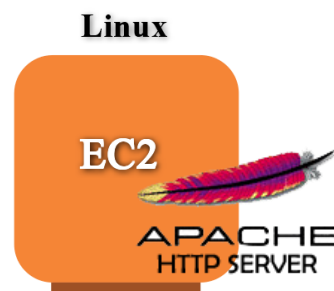
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**Configure Apache Web Server on
Linux Instance in AWS**



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Perundurai.**



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Used Services in AWS



Amazon EC2

Used Services in Linux Server

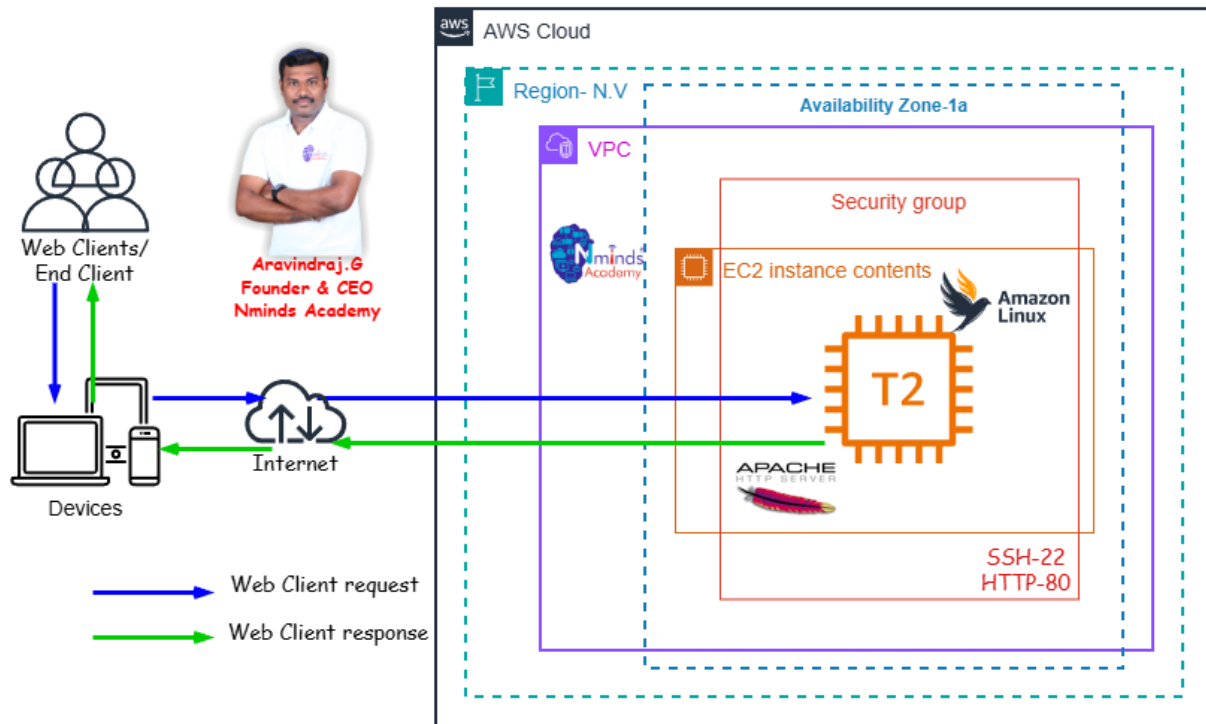


Apache Webserver



Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:

The screenshot displays the AWS Management Console interface for EC2 instances. The left sidebar shows navigation options like Dashboard, AWS Global View, Events, and various instance management tools. The main content area shows a list of instances, with one instance selected for detailed viewing.

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
linux	i-071cf392ee5414cd4	Running	t3.micro	Initializing	View alarms +	us-east-1d	ec2-54-209-104-139.co...	54.209

i-071cf392ee5414cd4 (linux)

Instance summary

Instance ID: i-071cf392ee5414cd4

Public IPv4 address: 54.209.104.139 | [open address](#)

Private IPv4 addresses: 172.31.40.156

Instance state: **Running**

Public DNS: ec2-54-209-104-139.compute-1.amazonaws.com | [open address](#)

IPV6 address: -

Hostname type: IP name: ip-172-31-40-156.ec2.internal

Private IP DNS name (IPv4 only): ip-172-31-40-156.ec2.internal

```
[ec2-user@ip-172-31-40-156 ~]$ sudo -i
[root@ip-172-31-40-156 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-40-156 ~]# yum install httpd -y
Last metadata expiration check: 0:00:09 ago on Mon May 11 17:00:18 2026.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
httpd                  x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      47 k
Installing dependencies:
apr                    x86_64            1.7.5-1.amzn2023.0.4  amazonlinux      129 k
apr-util               x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      97 k
apr-util-ldap          x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      13 k
generic-logos-httpd    noarch            18.0.0-12.amzn2023.0.3  amazonlinux      19 k
httpd-core              x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      1.4 M
httpd-filesystem        noarch            2.4.66-1.amzn2023.0.1  amazonlinux      13 k
httpd-tools             x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      81 k
libbrotli               x86_64            1.0.9-4.amzn2023.0.2  amazonlinux      315 k
mailcap                 noarch            2.1.49-3.amzn2023.0.3  amazonlinux      33 k
Installing weak dependencies:
apr-util-openssl        x86_64            1.6.3-1.amzn2023.0.2  amazonlinux      15 k
mod_http2                x86_64            2.0.27-1.amzn2023.0.3  amazonlinux      166 k
mod_lua                  x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      68 k
Transaction Summary
=====
Install 13 Packages

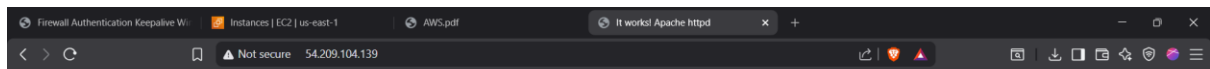
Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                3.7 MB/s | 129 kB | 00:00
(2/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm      360 kB/s | 13 kB | 00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm           2.4 MB/s | 97 kB | 00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm   659 kB/s | 15 kB | 00:00
(5/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm             2.0 MB/s | 47 kB | 00:00

Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64              6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64        7/13
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch   8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64       9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64          10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch            11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64         12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64           13/13
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64                               apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64                   apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch           httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64                     httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                     libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch                         mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64                         mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-40-156 ~]# systemctl start httpd
[root@ip-172-31-40-156 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Mon 2026-05-11 17:01:30 UTC; 9s ago
     Docs: man:httpd.service(8)
   Main PID: 26237 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1014)
    Memory: 13.5M
       CPU: 92ms
    CGroup: /system.slice/httpd.service
            └─26237 /usr/sbin/httpd -DFOREGROUND
              └─26262 /usr/sbin/httpd -DFOREGROUND
                └─26263 /usr/sbin/httpd -DFOREGROUND
                  └─26287 /usr/sbin/httpd -DFOREGROUND
                    └─26299 /usr/sbin/httpd -DFOREGROUND

May 11 17:01:30 ip-172-31-40-156.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
May 11 17:01:30 ip-172-31-40-156.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
May 11 17:01:30 ip-172-31-40-156.ec2.internal httpd[26237]: Server configured, listening on: port 80
[root@ip-172-31-40-156 ~]#
```





It works!

